HOLLOWAY ROAD EXTENSION - STEEL FRAME C - ANALYSIS DATA SHEET (sheet 1 of 5)

Structure. The building has three parallel plane frames on frame lines B, C and D (sheets 3 and 4), each of three bays (grid lines 1 to 4) and four storeys (levels G, L1, L2, L3 and roof R). Only frame C is analysed. Floors and roof span one way between the frame lines, so frame C carries the floor and roof loads of a strip extending from frame line C half-way to frame line B and half-way to frame line D (the frame C load strip). Ties on grid lines 1 to 4 carry no floor load and are not part of the frame C analysis.

Drawings. Sheet 3: framing plans of L1 and L2. Sheet 4: framing plans of L3 and roof. Sheet 5: column schedule, bracing elevations of lines C and D with storey heights, and the wind on frame C. On a framing plan a beam label is written directly above the beam it describes. Column sections, splice positions and base types are given in the column schedule; a column mark is frame line / grid line (C/2 is the column on frame line C at grid 2).

Analysis model.
1. Linear-elastic, first-order plane frame analysis of frame C in the state at the time stated in the site log (sheet 2). One load case: the loads below, with no load factors.
2. Centreline model: columns on their grid lines, beams on their level lines; bay widths from the framing plans, storey heights from sheet 5. Joint size and connection eccentricity are ignored.
3. E = 210 GPa for every member. Axial and bending deformation are included; shear deformation is ignored. Bending is about the major axis of every beam and column.
4. Columns are continuous through every floor. Each column has one splice, at the height given in the column schedule. The section changes at the splice; the splice itself is a full-strength rigid joint.
5. Beam-to-column connections are rigid (moment-resisting) unless shown as simple on the framing plan. A simple connection transmits no moment: that beam end is pinned to the column, and the column stays continuous.
6. Column bases are fixed or pinned as given in the column schedule.
7. Braces are pin-ended and carry axial force only. Brace self-weight is ignored.

Sections (A in cm2, I major axis in cm4, mass in kg/m):

203x203x46 UC	A 58.7	I 4568	46.1	406x178x54 UB	A 69.0	I 18720	54.1
203x203x52 UC	A 66.3	I 5259	52.0	406x178x60 UB	A 76.5	I 21600	60.1
203x203x60 UC	A 76.4	I 6125	60.0	457x191x67 UB	A 85.5	I 29380	67.1
254x254x73 UC	A 93.1	I 11410	73.1	457x191x74 UB	A 94.6	I 33320	74.3
254x254x89 UC	A 113.0	I 14270	88.9	533x210x82 UB	A 105.0	I 47540	82.2
168.3x6.3 CHS	A 32.1	I 1053	25.2	533x210x92 UB	A 117.0	I 55230	92.1
139.7x6.3 CHS	A 26.4	I 589	20.7	254x146x31 UB	A 39.7	I 4413	31.1

Loads on frame C.
1. Self-weight of every beam and column of frame C: mass x 9.81 N/kg, a uniform load along the member (vertical on beams, along the axis of columns).
2. Floor and roof area loads (kN/m2, given on each framing plan) act on the full span of every frame C beam at that level: line load = area load x width of the frame C load strip.
3. A storage zone adds its area load x width of the frame C load strip as a uniform line load over the zone length only, positioned by the dimensions on the framing plan.
4. Plant standing on a frame C beam is a vertical point load on that beam at the position the plant occupies, located by the dimensions on the roof plan.
5. Wind (sheet 5) is a set of horizontal point loads at the grid 1 joints of frame C, acting towards grid 4.
6. Temporary props are included only while they are installed.

Reporting conventions.
a. Horizontal displacement of a joint: in mm, positive towards grid 4.
b. Base reaction: vertical, in kN, upwards positive.
c. Beam end moment: the hogging moment at the beam end on the column centreline, in kNm.
d. Beam maximum sagging moment: the largest sagging moment anywhere along that beam, in kNm.
e. Column moment: the bending moment at the stated position on the column centreline, in kNm, positive when the face on the grid 4 side of the column (the side away from grid 1) is in tension.
f. Brace axial force: in kN, tension positive.

HOLLOWAY ROAD EXTENSION - FRAMES B, C, D - SITE TEMPORARY WORKS LOG (sheet 2 of 5)

Day 1 07:30 Frames B, C and D erected to roof level. Members, splices, braces and connections as drawn on sheets 3 to 5. All bolts tightened and checked except where recorded below.

Day 2 09:10 Temporary prop TP-1 installed under the L3 beam of frame C in bay 2-3 at midspan, for decking works.

Day 3 14:00 Decking complete at L1 and L2. Floor finishes and services loads now present as on the framing plans.

Day 4 08:20 Brace BR2 (frame C) disconnected at its upper end (grid 2, L2) to pass the main supply duct. Permit TW-12 open. Reinstatement after the duct is fixed, planned Day 14.

Day 4 15:40 Brace BR3 (frame D) checked after the duct route survey: in service, no action.

Day 5 11:45 Snagging, L2 beam of frame C in bay 3-4: end plate at grid 4 found with bolts not installed (holes misaligned). Until the bolts are installed this beam end acts as a simple connection. Grid 3 end of the same beam is complete.

Day 5 16:30 Snagging, L2 beam of frame D in bay 1-2: end plate at grid 1 re-bolted and checked; rigid as drawn.

Day 6 10:30 L3 beam of frame C in bay 1-2 at grid 1: simple connection checked against the L3 framing plan, as drawn. No action.

Day 7 15:20 Decking complete at L3 and roof. Floor finishes, roof finishes and services loads now present as on the framing plans.

Day 8 09:00 Storage pallets placed in storage zone S on L2 (sheet 3).

Day 9 13:40 AHU-2 lifted to the roof and landed on the frame C roof beam at temporary position T in bay 3-4 (sheet 4). Move to final position F planned for Day 15 after the plinth is cast.

Day 9 14:25 AHU-1 lifted to the roof and placed on its plinth on the frame D roof beam (sheet 4).

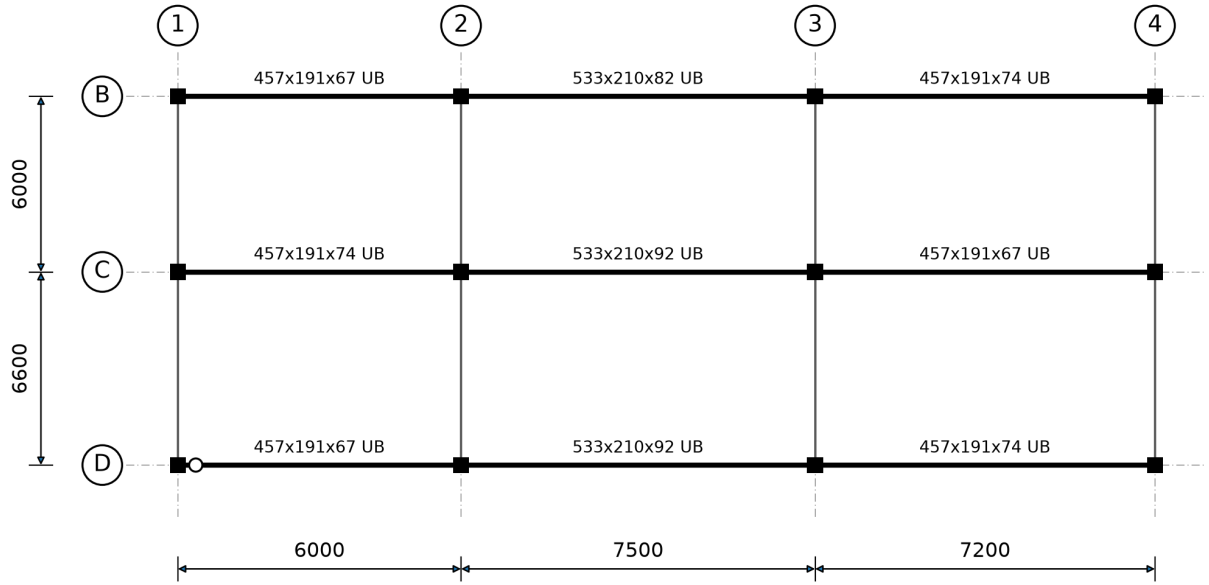
Day 9 16:10 Replacement bolts for the frame C L2 bay 3-4 end plate at grid 4 delivered. Installation booked for Day 13.

Day 10 08:00 Temporary prop TP-1 removed (decking cured). Permit TW-09 closed.

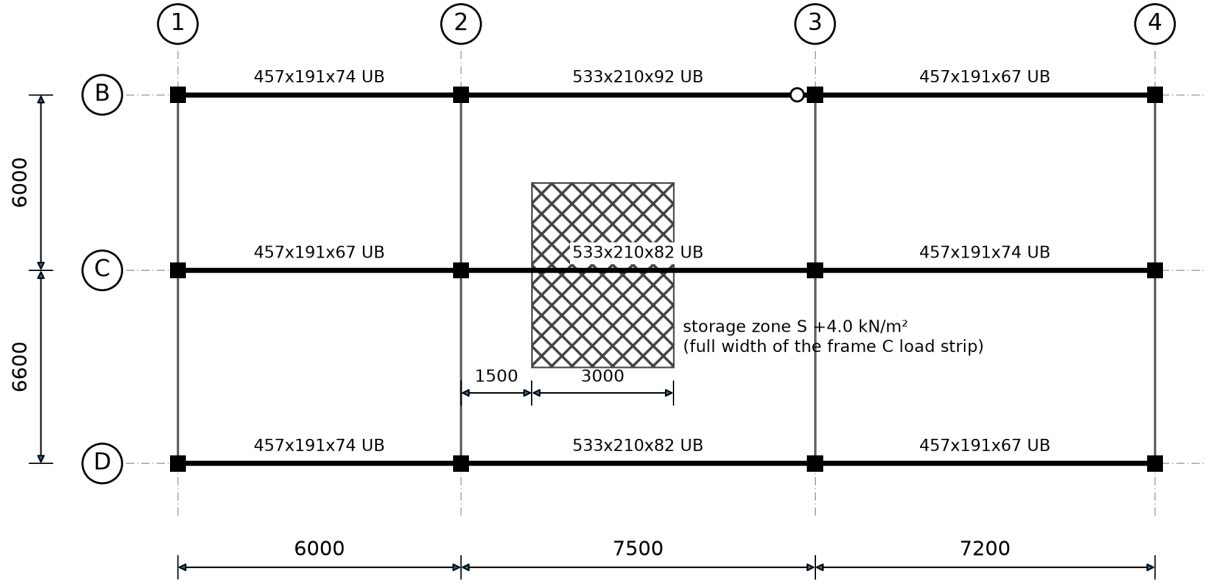
Day 11 17:30 Brace BR1 (frame C) inspected, bolts checked, in service.

Day 12 07:00 Engineer to analyse frame C for the state at this time, with the wind on frame C (sheet 5) acting.

FRAMING PLANS - LEVEL L1 AND LEVEL L2



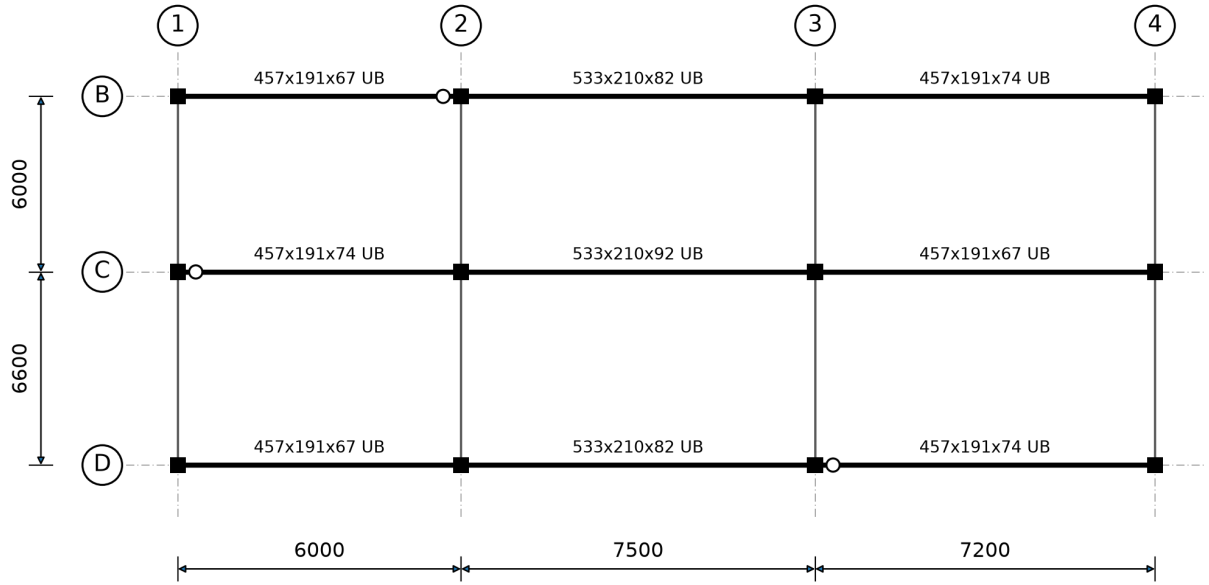
LEVEL L1 FRAMING PLAN floor load 7.5 kN/m² on all bays ties on grid lines 1-4: 254x146x31 UB



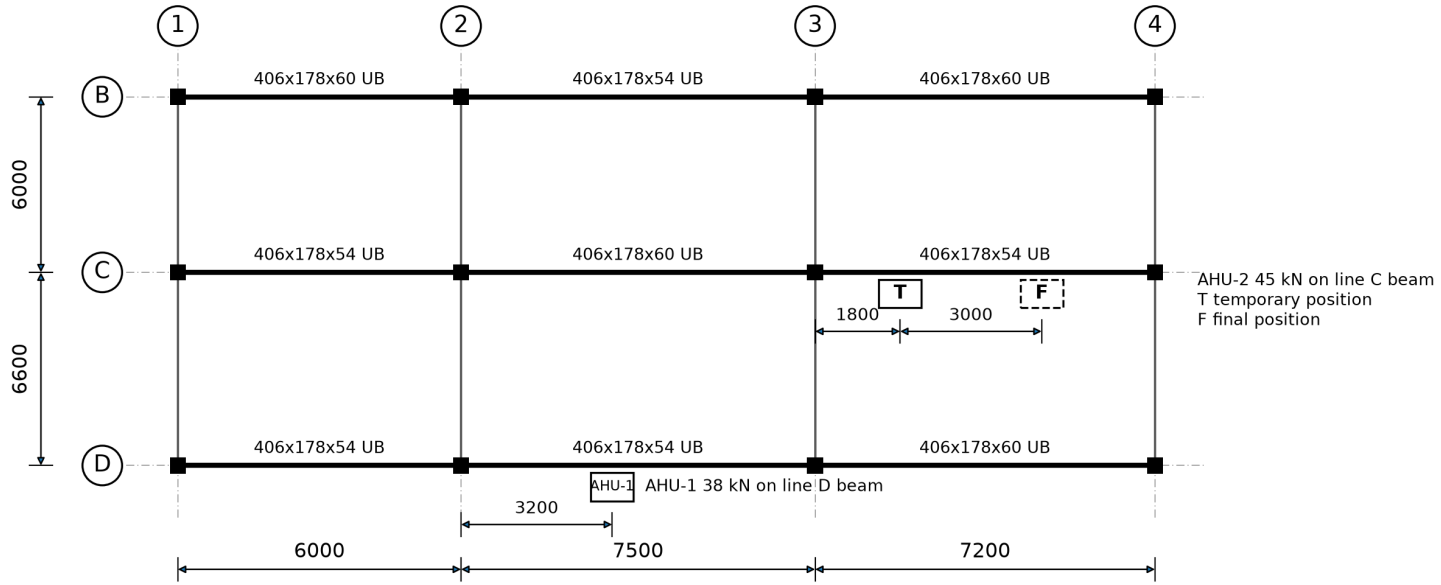
LEVEL L2 FRAMING PLAN floor load 7.5 kN/m² on all bays ties on grid lines 1-4: 254x146x31 UB

Legend: ■ column ○ simple connection at this beam end (other beam ends rigid) beam labels are written above the beam they describe

FRAMING PLANS - LEVEL L3 AND ROOF



LEVEL L3 FRAMING PLAN floor load 7.0 kN/m² on all bays ties on grid lines 1-4: 254x146x31 UB



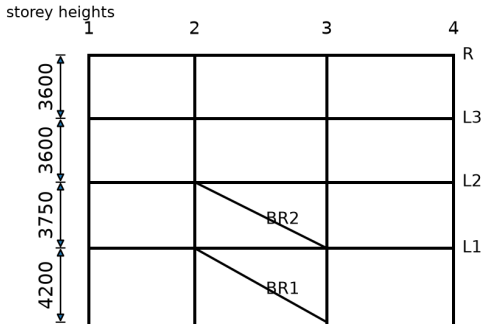
ROOF FRAMING PLAN floor load 5.0 kN/m² on all bays ties on grid lines 1-4: 254x146x31 UB

Legend: ■ column ○ simple connection at this beam end (other beam ends rigid) beam labels are written above the beam they describe

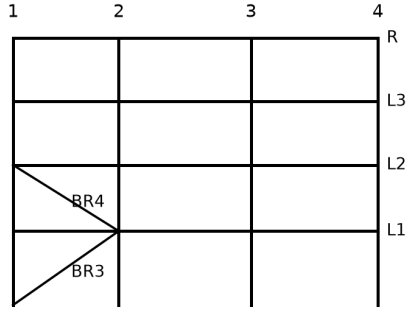
COLUMN SCHEDULE, BRACING ELEVATIONS AND WIND

Column	Below splice	Splice	Above splice	Base
B/1	203x203x52 UC	L2 + 600	203x203x46 UC	pinned
B/2	254x254x89 UC	L2 + 600	254x254x73 UC	pinned
B/3	254x254x73 UC	L1 + 600	203x203x60 UC	fixed
B/4	203x203x60 UC	L1 + 600	203x203x52 UC	pinned
C/1	203x203x60 UC	L2 + 600	203x203x52 UC	pinned
C/2	254x254x89 UC	L1 + 600	254x254x73 UC	fixed
C/3	254x254x89 UC	L2 + 600	254x254x73 UC	fixed
C/4	203x203x60 UC	L2 + 600	203x203x52 UC	pinned
D/1	203x203x60 UC	L1 + 600	203x203x52 UC	fixed
D/2	254x254x73 UC	L2 + 600	254x254x73 UC	fixed
D/3	254x254x89 UC	L1 + 600	254x254x73 UC	pinned
D/4	203x203x52 UC	L2 + 600	203x203x46 UC	pinned

Splice: height above the level shown, on the column centreline.
Column mark = frame line / grid line.



ELEVATION ON LINE C
braces 168.3x6.3 CHS, pin-ended



ELEVATION ON LINE D
braces 139.7x6.3 CHS, pin-ended

WIND ON FRAME C
horizontal point loads at the grid 1 joints of line C, acting towards grid 4

L1: 36 kN L2: 34 kN L3: 32 kN roof: 17 kN

Bracing elevations show bracing and levels only; member sizes are on the framing plans and the column schedule.